

Practice Problem 14: Gini Coefficient Analysis

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Problem Statement

Problem 14: Gini Coefficient

The Gini coefficient is a measure of statistical dispersion, often used for economic inequality. For a continuous distribution with CDF $F(x)$ and mean μ , it can be defined as:

$$G = \frac{1}{2\mu} \iint_{-\infty}^{\infty} |x - y| dF(x) dF(y)$$

Part 1: For the Pareto distribution with CDF $F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha$ for $x \geq x_m > 0$, $\alpha > 1$, show that the Gini coefficient is $G = \frac{1}{2\alpha-1}$.

Part 2: For a finite sample $\{x_1, \dots, x_n\}$, the Gini coefficient can be estimated by:

$$\hat{G} = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x}}$$

Compute $\hat{G}$ for the dataset $\{1, 3, 6, 10\}$.

Part 3: Prove that for any non-negative dataset, $0 \leq \hat{G} \leq 1$.

1 Solution to Part 1: Gini Coefficient for Pareto Distribution

1.1 Step 1: Derive the PDF from the CDF

The Pareto distribution has the cumulative distribution function (CDF):

$$F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha \quad \text{for } x \geq x_m > 0, \quad \alpha > 1$$

The probability density function (PDF) is obtained by differentiating the CDF:

$$f(x) = \frac{d}{dx} F(x) = \frac{d}{dx} \left[1 - \left(\frac{x_m}{x}\right)^\alpha \right]$$

Using the chain rule:

$$f(x) = -\frac{d}{dx} [x_m^\alpha x^{-\alpha}] = -x_m^\alpha \cdot (-\alpha) x^{-\alpha-1} = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$

1.2 Step 2: Calculate the Mean of the Pareto Distribution

The mean is computed as:

$$\mu = \mathbb{E}[X] = \int_{x_m}^{\infty} x \cdot f(x) dx = \int_{x_m}^{\infty} x \cdot \frac{\alpha x_m^\alpha}{x^{\alpha+1}} dx$$

Simplifying:

$$\mu = \alpha x_m^\alpha \int_{x_m}^{\infty} x^{-\alpha} dx$$

Evaluating the integral:

$$\int_{x_m}^{\infty} x^{-\alpha} dx = \left[\frac{x^{1-\alpha}}{1-\alpha} \right]_{x_m}^{\infty} = \lim_{x \rightarrow \infty} \frac{x^{1-\alpha}}{1-\alpha} - \frac{x_m^{1-\alpha}}{1-\alpha}$$

Since $\alpha > 1$, we have $1 - \alpha < 0$, so $\lim_{x \rightarrow \infty} x^{1-\alpha} = 0$:

$$\int_{x_m}^{\infty} x^{-\alpha} dx = 0 - \frac{x_m^{1-\alpha}}{1-\alpha} = \frac{x_m^{1-\alpha}}{\alpha-1}$$

Therefore, the mean is:

$$\mu = \alpha x_m^\alpha \cdot \frac{x_m^{1-\alpha}}{\alpha-1} = \frac{\alpha x_m}{\alpha-1}$$

1.3 Step 3: Simplify the Gini Coefficient Formula

The Gini coefficient is defined as:

$$G = \frac{1}{2\mu} \iint_{-\infty}^{\infty} |x - y| dF(x) dF(y)$$

Using symmetry, we can rewrite this as:

$$G = \frac{1}{\mu} \iint_{y < x} (x - y) dF(x) dF(y)$$

This becomes:

$$G = \frac{1}{\mu} \int_{x_m}^{\infty} \left[\int_{x_m}^x (x - y) f(y) dy \right] f(x) dx$$

1.4 Step 4: Calculate the Inner Integral $J(x)$

Define:

$$J(x) = \int_{x_m}^x (x - y)f(y) dy = \int_{x_m}^x (x - y) \cdot \frac{\alpha x_m^\alpha}{y^{\alpha+1}} dy$$

Factor out constants:

$$J(x) = \alpha x_m^\alpha \int_{x_m}^x \left(\frac{x}{y^{\alpha+1}} - \frac{1}{y^\alpha} \right) dy$$

Split into two integrals:

$$J(x) = \alpha x_m^\alpha \left[x \int_{x_m}^x y^{-\alpha-1} dy - \int_{x_m}^x y^{-\alpha} dy \right]$$

First integral:

$$\int_{x_m}^x y^{-\alpha-1} dy = \left[\frac{y^{-\alpha}}{-\alpha} \right]_{x_m}^x = -\frac{1}{\alpha} (x^{-\alpha} - x_m^{-\alpha}) = \frac{1}{\alpha} (x_m^{-\alpha} - x^{-\alpha})$$

Second integral:

$$\int_{x_m}^x y^{-\alpha} dy = \left[\frac{y^{1-\alpha}}{1-\alpha} \right]_{x_m}^x = \frac{1}{1-\alpha} (x^{1-\alpha} - x_m^{1-\alpha}) = \frac{1}{\alpha-1} (x_m^{1-\alpha} - x^{1-\alpha})$$

Substituting back:

$$J(x) = \alpha x_m^\alpha \left[\frac{x}{\alpha} (x_m^{-\alpha} - x^{-\alpha}) - \frac{1}{\alpha-1} (x_m^{1-\alpha} - x^{1-\alpha}) \right]$$

Simplifying:

$$J(x) = x_m^\alpha \left[x (x_m^{-\alpha} - x^{-\alpha}) - \frac{\alpha}{\alpha-1} (x_m^{1-\alpha} - x^{1-\alpha}) \right]$$

$$J(x) = x - x \cdot x_m^\alpha x^{-\alpha} - \frac{\alpha}{\alpha-1} x_m + \frac{\alpha}{\alpha-1} x \cdot x_m^\alpha x^{-\alpha}$$

$$J(x) = x \left(1 - \left(\frac{x_m}{x} \right)^\alpha \right) - \frac{\alpha}{\alpha-1} x_m + \frac{\alpha}{\alpha-1} x \left(\frac{x_m}{x} \right)^\alpha$$

1.5 Step 5: Calculate the Outer Integral

Now we compute:

$$I = \int_{x_m}^\infty J(x)f(x) dx = \int_{x_m}^\infty J(x) \cdot \frac{\alpha x_m^\alpha}{x^{\alpha+1}} dx$$

Substituting $J(x)$:

$$I = \alpha x_m^\alpha \int_{x_m}^\infty \left[x \left(1 - \left(\frac{x_m}{x} \right)^\alpha \right) - \frac{\alpha}{\alpha-1} x_m + \frac{\alpha}{\alpha-1} x \left(\frac{x_m}{x} \right)^\alpha \right] \frac{1}{x^{\alpha+1}} dx$$

Expanding:

$$I = \alpha x_m^\alpha \int_{x_m}^{\infty} \left[x^{-\alpha} - x_m^\alpha x^{-2\alpha} - \frac{\alpha}{\alpha-1} x_m x^{-\alpha-1} + \frac{\alpha}{\alpha-1} x_m^\alpha x^{-2\alpha} \right] dx$$

Combining the $x^{-2\alpha}$ terms:

$$\left(\frac{\alpha}{\alpha-1} - 1 \right) x_m^\alpha x^{-2\alpha} = \frac{\alpha - (\alpha-1)}{\alpha-1} x_m^\alpha x^{-2\alpha} = \frac{1}{\alpha-1} x_m^\alpha x^{-2\alpha}$$

So:

$$I = \alpha x_m^\alpha \int_{x_m}^{\infty} \left[x^{-\alpha} - \frac{\alpha}{\alpha-1} x_m x^{-\alpha-1} + \frac{1}{\alpha-1} x_m^\alpha x^{-2\alpha} \right] dx$$

1.6 Step 6: Evaluate Each Component Integral

First integral:

$$I_1 = \int_{x_m}^{\infty} x^{-\alpha} dx = \left[\frac{x^{1-\alpha}}{1-\alpha} \right]_{x_m}^{\infty} = 0 - \frac{x_m^{1-\alpha}}{1-\alpha} = \frac{x_m^{1-\alpha}}{\alpha-1}$$

Second integral:

$$I_2 = \int_{x_m}^{\infty} x^{-\alpha-1} dx = \left[\frac{x^{-\alpha}}{-\alpha} \right]_{x_m}^{\infty} = 0 - \frac{x_m^{-\alpha}}{-\alpha} = \frac{x_m^{-\alpha}}{\alpha}$$

Third integral:

$$I_3 = \int_{x_m}^{\infty} x^{-2\alpha} dx = \left[\frac{x^{1-2\alpha}}{1-2\alpha} \right]_{x_m}^{\infty} = 0 - \frac{x_m^{1-2\alpha}}{1-2\alpha} = \frac{x_m^{1-2\alpha}}{2\alpha-1}$$

Note: Since $\alpha > 1$, we have $2\alpha > 2 > 1$, so $1 - 2\alpha < 0$, ensuring convergence.

1.7 Step 7: Combine All Terms

Substituting back:

$$I = \alpha x_m^\alpha \left[I_1 - \frac{\alpha}{\alpha-1} x_m I_2 + \frac{1}{\alpha-1} x_m^\alpha I_3 \right]$$

$$I = \alpha x_m^\alpha \left[\frac{x_m^{1-\alpha}}{\alpha-1} - \frac{\alpha}{\alpha-1} x_m \cdot \frac{x_m^{-\alpha}}{\alpha} + \frac{1}{\alpha-1} x_m^\alpha \cdot \frac{x_m^{1-2\alpha}}{2\alpha-1} \right]$$

Simplifying each term:

First term:

$$\alpha x_m^\alpha \cdot \frac{x_m^{1-\alpha}}{\alpha-1} = \frac{\alpha x_m}{\alpha-1}$$

Second term:

$$\alpha x_m^\alpha \cdot \frac{\alpha}{\alpha-1} x_m \cdot \frac{x_m^{-\alpha}}{\alpha} = \frac{\alpha x_m^\alpha \cdot x_m \cdot x_m^{-\alpha}}{\alpha-1} = \frac{\alpha x_m}{\alpha-1}$$

Third term:

$$\alpha x_m^\alpha \cdot \frac{1}{\alpha-1} x_m^\alpha \cdot \frac{x_m^{1-2\alpha}}{2\alpha-1} = \frac{\alpha x_m^{2\alpha} \cdot x_m^{1-2\alpha}}{(\alpha-1)(2\alpha-1)} = \frac{\alpha x_m}{(\alpha-1)(2\alpha-1)}$$

Therefore:

$$I = \frac{\alpha x_m}{\alpha-1} - \frac{\alpha x_m}{\alpha-1} + \frac{\alpha x_m}{(\alpha-1)(2\alpha-1)} = \frac{\alpha x_m}{(\alpha-1)(2\alpha-1)}$$

1.8 Step 8: Calculate the Gini Coefficient

The Gini coefficient is:

$$G = \frac{I}{\mu} = \frac{\frac{\alpha x_m}{(\alpha-1)(2\alpha-1)}}{\frac{\alpha x_m}{\alpha-1}} = \frac{\alpha x_m}{(\alpha-1)(2\alpha-1)} \cdot \frac{\alpha-1}{\alpha x_m}$$

$$G = \frac{1}{2\alpha-1}$$

Result for Part 1

$$G = \frac{1}{2\alpha-1}$$

This elegant result shows that the Gini coefficient for a Pareto distribution depends only on the shape parameter α . As α increases, the inequality (measured by G) decreases. For example:

- When $\alpha = 1.5$: $G = \frac{1}{2} = 0.5$ (high inequality)
- When $\alpha = 2$: $G = \frac{1}{3} \approx 0.333$ (moderate inequality)
- When $\alpha = 5$: $G = \frac{1}{9} \approx 0.111$ (low inequality)

2 Solution to Part 2: Empirical Gini Coefficient

2.1 Step 1: Define the Estimator and Dataset

For a finite sample $\{x_1, \dots, x_n\}$, the Gini coefficient is estimated by:

$$\hat{G} = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x}}$$

Given dataset: $\{1, 3, 6, 10\}$, so $n = 4$.

2.2 Step 2: Calculate the Sample Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1 + 3 + 6 + 10}{4} = \frac{20}{4} = 5$$

2.3 Step 3: Compute All Pairwise Absolute Differences

We need to calculate $|x_i - x_j|$ for all pairs (i, j) . Let's create a difference matrix:

Detailed calculations for each row:

$ x_i - x_j $	$x_1 = 1$	$x_2 = 3$	$x_3 = 6$	$x_4 = 10$	Row Sum
$x_1 = 1$	0	2	5	9	16
$x_2 = 3$	2	0	3	7	12
$x_3 = 6$	5	3	0	4	12
$x_4 = 10$	9	7	4	0	20
Total					60

Table 1: Pairwise absolute differences matrix

Row 1 ($x_1 = 1$):

$$|1 - 1| = 0$$

$$|1 - 3| = |-2| = 2$$

$$|1 - 6| = |-5| = 5$$

$$|1 - 10| = |-9| = 9$$

$$\text{Sum} = 0 + 2 + 5 + 9 = 16$$

Row 2 ($x_2 = 3$):

$$|3 - 1| = 2$$

$$|3 - 3| = 0$$

$$|3 - 6| = |-3| = 3$$

$$|3 - 10| = |-7| = 7$$

$$\text{Sum} = 2 + 0 + 3 + 7 = 12$$

Row 3 ($x_3 = 6$):

$$|6 - 1| = 5$$

$$|6 - 3| = 3$$

$$|6 - 6| = 0$$

$$|6 - 10| = |-4| = 4$$

$$\text{Sum} = 5 + 3 + 0 + 4 = 12$$

Row 4 ($x_4 = 10$):

$$|10 - 1| = 9$$

$$|10 - 3| = 7$$

$$|10 - 6| = 4$$

$$|10 - 10| = 0$$

$$\text{Sum} = 9 + 7 + 4 + 0 = 20$$

Grand total:

$$S = \sum_{i=1}^4 \sum_{j=1}^4 |x_i - x_j| = 16 + 12 + 12 + 20 = 60$$

2.4 Step 4: Calculate the Gini Coefficient

Now we can compute $\hat{G}$:

$$\hat{G} = \frac{S}{2n^2\bar{x}} = \frac{60}{2 \times 4^2 \times 5} = \frac{60}{2 \times 16 \times 5} = \frac{60}{160} = \frac{3}{8} = 0.375$$

Result for Part 2

$$\hat{G} = 0.375 = \frac{3}{8}$$

Interpretation: A Gini coefficient of 0.375 indicates moderate inequality in the dataset. This value is between 0 (perfect equality) and 1 (perfect inequality). The dataset $\{1, 3, 6, 10\}$ shows some dispersion, with the largest value being 10 times the smallest value.

3 Solution to Part 3: Bounds on the Gini Coefficient

3.1 Proof that $0 \leq \hat{G} \leq 1$ for Non-Negative Data

Theorem: For any non-negative dataset $\{x_1, x_2, \dots, x_n\}$ with $x_i \geq 0$ for all i , the empirical Gini coefficient satisfies:

$$0 \leq \hat{G} \leq 1$$

3.2 Part A: Lower Bound ($\hat{G} \geq 0$)

The Gini coefficient is defined as:

$$\hat{G} = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2\bar{x}}$$

Step 1: Since absolute values are always non-negative:

$$|x_i - x_j| \geq 0 \quad \text{for all } i, j$$

Therefore:

$$\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \geq 0$$

Step 2: The denominator is positive when at least one $x_i > 0$ (otherwise $\bar{x} = 0$ and $\hat{G}$ is undefined):

$$2n^2\bar{x} = 2n^2 \cdot \frac{\sum_{i=1}^n x_i}{n} = 2n \sum_{i=1}^n x_i > 0$$

Step 3: Since both numerator and denominator are non-negative:

$$\hat{G} = \frac{(\text{non-negative})}{(\text{positive})} \geq 0$$

When does $\hat{G} = 0$? Equality holds when:

$$\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| = 0$$

This occurs if and only if $x_i = x_j$ for all i, j , meaning all values are equal (perfect equality).

3.3 Part B: Upper Bound ($\hat{G} \leq 1$)

Key inequality: For non-negative values $x_i, x_j \geq 0$, we have:

$$|x_i - x_j| \leq x_i + x_j$$

Proof of key inequality: Without loss of generality, assume $x_i \geq x_j \geq 0$. Then:

$$\begin{aligned} |x_i - x_j| &= x_i - x_j \\ &\leq x_i - x_j + 2x_j \quad (\text{since } x_j \geq 0) \\ &= x_i + x_j \end{aligned}$$

Step 1: Applying this inequality to the double sum:

$$\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \leq \sum_{i=1}^n \sum_{j=1}^n (x_i + x_j)$$

Step 2: Expand the right-hand side:

$$\sum_{i=1}^n \sum_{j=1}^n (x_i + x_j) = \sum_{i=1}^n \sum_{j=1}^n x_i + \sum_{i=1}^n \sum_{j=1}^n x_j$$

For the first term, x_i is constant with respect to the inner sum over j :

$$\sum_{i=1}^n \sum_{j=1}^n x_i = \sum_{i=1}^n x_i \sum_{j=1}^n 1 = \sum_{i=1}^n x_i \cdot n = n \sum_{i=1}^n x_i$$

Similarly for the second term:

$$\sum_{i=1}^n \sum_{j=1}^n x_j = \sum_{i=1}^n \left(\sum_{j=1}^n x_j \right) = n \sum_{j=1}^n x_j$$

Therefore:

$$\sum_{i=1}^n \sum_{j=1}^n (x_i + x_j) = n \sum_{i=1}^n x_i + n \sum_{j=1}^n x_j = 2n \sum_{i=1}^n x_i$$

Step 3: Substitute into the Gini formula:

$$\hat{G} = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x}} \leq \frac{2n \sum_{i=1}^n x_i}{2n^2 \bar{x}}$$

Step 4: Recall that $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, so:

$$\hat{G} \leq \frac{2n \sum_{i=1}^n x_i}{2n^2 \cdot \frac{1}{n} \sum_{i=1}^n x_i} = \frac{2n \sum_{i=1}^n x_i}{2n \sum_{i=1}^n x_i} = 1$$

3.4 When does $\hat{G} = 1$?

Maximum inequality occurs when $|x_i - x_j| = x_i + x_j$ for all pairs. From our inequality proof, this happens when one of the values equals zero.

Example: Consider the dataset $\{0, 0, \dots, 0, a\}$ where $n - 1$ values are zero and one value is $a > 0$.

Then $\bar{x} = \frac{a}{n}$, and:

$$\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| = 2(n-1) \times a = 2(n-1)a$$

(Each of the $n - 1$ zeros paired with a contributes $|0 - a| = a$, counted twice due to symmetry)

Therefore:

$$\hat{G} = \frac{2(n-1)a}{2n^2 \cdot \frac{a}{n}} = \frac{2(n-1)a}{2na} = \frac{n-1}{n} = 1 - \frac{1}{n} \rightarrow 1 \text{ as } n \rightarrow \infty$$

For finite n , the maximum is $\frac{n-1}{n}$, which approaches 1 as the sample size increases.

Result for Part 3

Proven:

$$0 \leq \hat{G} \leq 1$$

Summary:

- $\hat{G} = 0$ occurs when all values are equal (perfect equality)
- $\hat{G} = 1$ (or close to 1) occurs when all wealth is concentrated in one observation (perfect inequality)
- The bounds hold for any non-negative dataset

4 Summary and Key Insights

Final Summary

Main Results

1. Theoretical Gini for Pareto Distribution:

$$G = \frac{1}{2\alpha - 1}$$

This shows that higher shape parameter α leads to lower inequality.

2. Empirical Gini for Dataset $\{1, 3, 6, 10\}$:

$$\hat{G} = 0.375$$

This indicates moderate inequality in the sample.

3. Universal Bounds:

$$0 \leq \hat{G} \leq 1 \quad \text{for all non-negative datasets}$$

Applications

The Gini coefficient is widely used in:

- **Economics:** Measuring income and wealth inequality
- **Healthcare:** Assessing disparities in health outcomes
- **Ecology:** Measuring biodiversity and species distribution
- **Machine Learning:** Feature selection and decision tree splitting criteria

Key Properties

1. **Scale invariance:** Multiplying all values by a constant doesn't change G
2. **Translation sensitivity:** Adding a constant to all values changes G
3. **Population independence:** For Pareto, G depends only on α , not x_m
4. **Interpretability:** Values between 0.25-0.35 indicate reasonable equality, above 0.50 indicates high inequality

Conclusion

This problem demonstrates the mathematical rigor required to analyze inequality measures. The Pareto distribution's analytical Gini coefficient provides a theoretical foundation, while the empirical estimate shows practical application. The proof of bounds ensures the measure is well-behaved and interpretable across all datasets.

The step-by-step derivations reveal the deep connections between probability theory, calculus, and statistical measurement, showcasing how abstract mathematical concepts translate into practical tools for understanding real-world phenomena.